Best Candidate Review

Project: AI-Enabled Underwater Garbage Detection Using Computer Vision

1. Objective

The purpose of this review was to compare the completed model experiments, identify the strongest candidate and determine the next direction for model evaluation.

2. Candidate comparison

The three completed V2.1 experiments were compared using validation metrics.

Metric	YOLO11s baseline	YOLO11s oversampled	YOLOv8s baseline
Precision	**0.519**	0.494	0.439
Recall	0.234	0.198	**0.238**
mAP@50	**0.236**	0.203	0.216
mAP@50–95	**0.143**	0.125	0.130
Inference time	3.9 ms	4.4 ms	3.7 ms
Parameters	9.42M	9.42M	11.13M
GFLOPs	21.5	21.5	28.5

3. Selected candidate

Selected model: YOLO11s baseline trained on V2.1

Reasons for selection:

- Highest validation precision among the compared candidates.
- Highest validation mAP@50 and mAP@50–95.
- Lower parameter count and GFLOPs than YOLOv8s.
- Better aggregate validation performance than the oversampled configuration.
- Evaluated using a dataset split with no source-video overlap.

4. Representative strengths

The model showed stronger validation performance for:

- animal_crab
- rov

- animal_fish
- animal_eel
- trash_metal
- trash_fabric

These results suggest that the model can detect several common and visually distinguishable categories reasonably well.

5. Representative limitations

The model showed weak validation performance for:

- plant
- animal_starfish
- animal_shells
- animal_etc
- trash_fishing_gear
- trash_paper
- trash_rubber
- trash_wood

The class-wise results highlight the need for better representation and generalization for underrepresented categories.

6. Final checkpoint

Selected checkpoint:

/content/drive/MyDrive/material_version/yolo_runs/V2.1_yolo11s_baseline_640/weights/best.pt

Configuration:

- Architecture: YOLO11s
- Input resolution: 640 × 640
- Training epochs: 100
- Dataset: V2.1
- Validation mAP@50: 0.236
- Validation mAP@50–95: 0.143

7. Approved next direction

The selected YOLO11s baseline should be retained as the reference candidate for subsequent evaluation.

The next steps should include:

1. Evaluate the selected checkpoint on the untouched V2.1 test set.
2. Inspect predictions and identify false positives and false negatives.
3. Investigate weak classes through class-wise error analysis.
4. Explore additional approaches to class imbalance without compromising the validation protocol.
5. Conduct dedicated robustness experiments under low visibility, occlusion and clutter.

Conclusion

YOLO11s baseline on V2.1 was selected as the best completed candidate based on aggregate validation metrics. Although it demonstrated stronger overall performance than the other completed V2.1 configurations, class-wise limitations remain. The final test evaluation is required before making claims about generalization to unseen data.